ADITYA SHARMA

AI/ML Engineer | Computer Vision Enthusiast | Python Developer

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ABOUT ME

Computer Science undergraduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Python Development, TensorFlow, PyTorch, and Full-Stack Application Development. Currently working as an AI/ML Intern developing image processing and computer vision solutions. Passionate about building scalable AI-powered products and solving real-world problems through data-driven technologies.

SKILLS

Languages: Python, C++, Java, SQL
AI/ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, CNNs, Deep Learning, Computer Vision
Databases: MySQL, SQLite
Backend: FastAPI, Flask, Node.js, REST APIs
Data Science: Pandas, NumPy, Matplotlib, Seaborn
Tools: Git, GitHub, Android Studio, VS Code, Jupyter Notebook

EXPERIENCE

AI/ML Intern, AiAssured TestAIng May 2026 – Present

- Developing and evaluating computer vision and image processing models for real-world AI applications, improving baseline accuracy by 12%.
- Working with Python, OpenCV, TensorFlow, and PyTorch to build and optimize machine learning pipelines, reducing inference time by 20%.
- Performing data preprocessing, model training, testing, and performance analysis on image datasets exceeding 10,000 samples.
- Contributing to AI-driven solutions focused on image classification, object detection, and visual intelligence systems.

PROJECTS

Dr Drive AI-Powered Vehicle Inspection & Predictive Maintenance Platform March 2026 – Present

- Developed a full-stack **Android application** to analyze vehicle health using On-Board Diagnostics (OBD) data and computer vision.
- Implemented AI models to detect mechanical faults and predict failures with 85% accuracy, estimating repair costs based on real-time data.
- Integrated image-based damage detection for dents, scratches, and structural issues using deep learning techniques.
- Built a smart valuation system utilizing a dataset of 50,000+ vehicle records to estimate fair resale price and detect potential listing fraud.

Traffic Signal Waiting Time Analyzer Feb 2026 – March 2026

- Engineered a C++ traffic simulation engine utilizing Queue data structures to model multi-lane intersections and track vehicle arrival times.
- Implemented and benchmarked scheduling algorithms (Fixed-Time, Greedy, and Priority Queue), applying Prefix Sum arrays to efficiently compute cumulative wait times.
- Developed a robust data pipeline connecting the C++ backend to a Python (Streamlit) frontend, parsing JSON outputs for real-time monitoring.

AI Powered Disaster Management Robot Sept 2025 – Nov 2025

- Designed an AI-based system to detect and locate people trapped under debris with a 92% success rate in simulated disaster scenarios.
- Trained ML models to accurately monitor **pulse, heartbeat, oxygen, and Carbon Dioxide (CO_2) levels** for real-time survivor detection.
- Integrated computer vision techniques for identifying human presence using advanced image detection algorithms in low-light environments.
- Focused on building a scalable and life-saving hardware-software solution for emergency response and search-and-rescue operations.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** Google Advanced Data Analytics, Google AI Essentials; HackerRank SQL Basic/Intermediate/Advanced; HackerRank Python Basic
- **Achievements:** Finalist, Statistella BASH 8.0 Data Analytics Competition; Rank 1, Tableau Dashboard Design Round 1

EDUCATION

Jaypee Institute of Information Technology 2024 – 2029
B.Tech + Integrated M.Tech in Computer Science CGPA: 7.56

Masai School IIT Patna Expected: 2026
Applied AI & ML Essentials

Delhi Public School Ghaziabad 2023
Senior Secondary (Class XII) (CBSE) Percentage: 92.2%